

CRUDE OIL MARKET REGIME PREDICTION

Hidden Markov Models, Bayesian Networks, PCA,
and Rolling-Window Validation

Technical Portfolio Report

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Portfolio presentation of an academic group project

Focus: probabilistic regime classification and model interpretation — not investment advice.

Executive Summary

This project develops a probabilistic framework for classifying the next-month crude-oil market regime. The implementation combines Hidden Markov Models (HMMs) for regime identification with Bayesian Networks (BNs) for next-month regime prediction. A baseline categorical pipeline is compared with a PCA-based Gaussian-HMM pipeline using chronological rolling-window validation.

Item	Portfolio finding
Objective	Predict the next-month crude-oil regime: Low, Middle, or High.
Data	Monthly market, macroeconomic, supply-demand, industrial, geopolitical, weather, and financial variables, 1997–2024.
Models	HMM regime detection + Bayesian Network inference; baseline and PCA-based pipelines.
BN structure search	Hill Climbing + BIC, Hill Climbing + BDeu, and PC; candidate selected by validation Macro F1.
Best validation pipeline	PCA-based pipeline with HC-BIC selected most often.
Key validation result	Validation accuracy increased from 0.755 (baseline) to 0.856 (PCA pipeline); Macro F1 improved only modestly, 0.710 to 0.724.
Main structural result	The current crude-oil regime is the dominant direct predictor of the next-month regime.
Final unseen period	12 evaluated months, all Middle regime; accuracy/F1 = 1.00 but no regime diversity.
Trading result	The rule High=long, Middle=flat, Low=short generated zero cumulative return because every prediction was Middle.

Key interpretation

The PCA-based pipeline improved average validation accuracy and weighted F1 while reducing network size, but the evidence does not establish that PCA alone caused the improvement. The two pipelines also differ in preprocessing and HMM specification. In addition, the final 12-month test contains only one realized class, so its 100% accuracy should not be treated as evidence of general multiclass superiority.

1. Problem Definition

Crude-oil markets are influenced by interacting economic, supply-demand, financial, geopolitical, and environmental factors. Instead of treating the market as a single stationary process, this project asks whether latent regimes can provide a more structured representation of market conditions and whether a Bayesian Network can predict the regime one month ahead.

The implemented prediction target is the next-month HMM-defined crude-oil regime. This is an important distinction from direct price forecasting: the model predicts a categorical state (Low, Middle, or High), and the final trading experiment maps those states into positions.

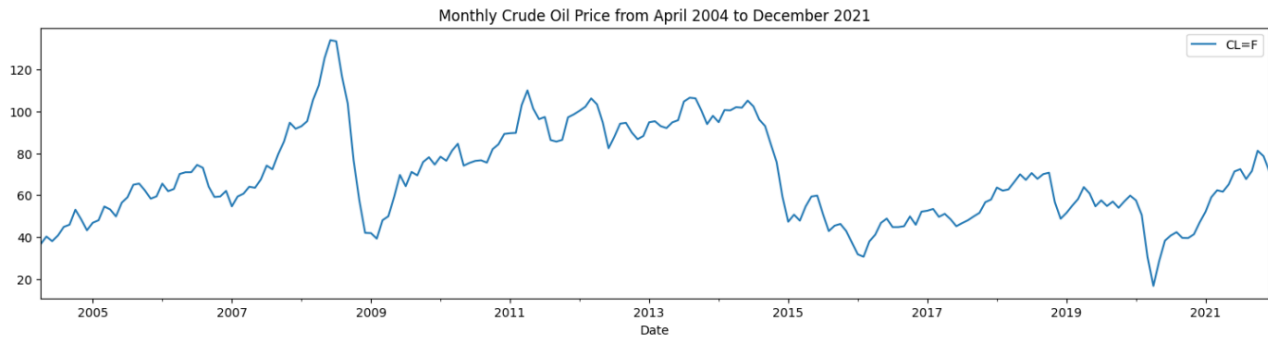


Figure 1. Monthly crude-oil futures price series used to motivate regime-based modelling.

Research Questions

- Can HMMs identify persistent and interpretable market regimes?
- Can Bayesian Networks predict the next-month crude-oil regime out of sample?
- Does the PCA-based pipeline improve predictive performance while reducing network complexity?
- Which variables or latent factors are most consistently connected to the next-month regime?
- Does the learned dependency structure change across pre-COVID, COVID-inclusive, and later periods?
- Can the predicted regimes be translated into a useful trading rule?

2. Data and Feature Design

The dataset combines monthly observations from December 1997 through March 2024. Sources used in the original project include the U.S. Energy Information Administration (EIA), FRED, Yahoo Finance, and the Geopolitical Risk dataset. The modeling dataset contains crude-oil market variables together with macroeconomic, supply-demand, industrial, financial, geopolitical, and weather indicators.

Category	Representative variables
Supply	OPEC production; non-OPEC production
Demand	OECD and non-OECD petroleum consumption
Industrial / macro	Industrial production indices for the U.S., OECD, China, India, and Russia
U.S. oil industry	Crude-oil industrial production, petroleum/coal inventories, capacity utilization
Financial	VIX, U.S. Dollar Index, NASDAQ Composite, U.S. Treasury yield
Other	Global temperature anomaly; Geopolitical Risk Index
Target input	Crude-oil futures (CL=F), transformed into HMM regimes

Data preparation principles

- Time-series ordering is preserved; random cross-validation is not used.
- Stationarity is assessed using ADF/KPSS diagnostics and variables are transformed where required.
- Missing-data treatment and transformations are applied before model construction.
- Outliers associated with major market events are retained when they appear economically meaningful rather than treated automatically as errors.
- For the PCA pipeline, scaling and PCA are fit on training data and then applied to validation data to reduce look-ahead bias.

Target definition

For each month t , the Bayesian Network target is the crude-oil regime at $t+1$. The current crude-oil regime at t remains available as a predictor. The model therefore estimates a one-step-ahead regime transition, not a dollar price forecast.

3. Methodology

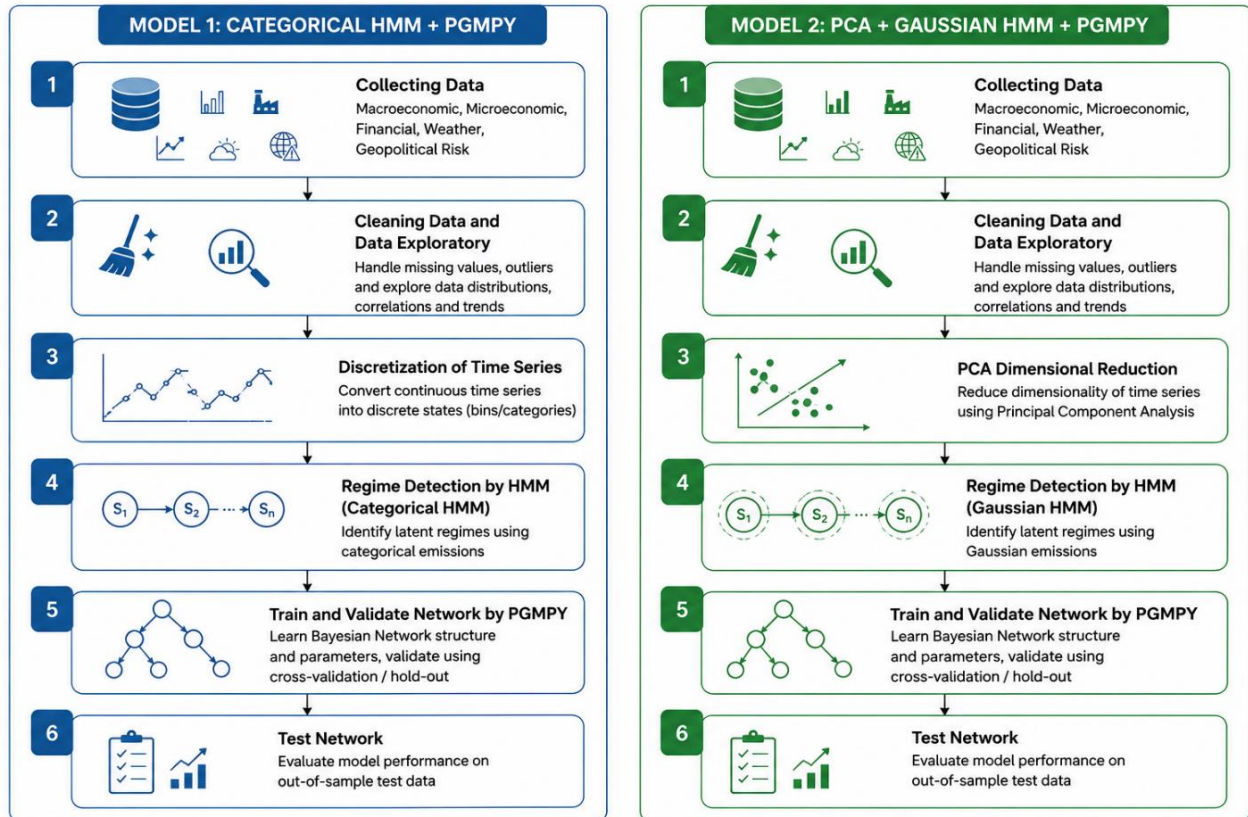


Figure 2. End-to-end comparison of the baseline categorical-HMM pipeline and the PCA + Gaussian-HMM pipeline.

3.1 Baseline pipeline

- Continuous variables are transformed and discretized using thresholds derived from training data.
- A three-state Categorical HMM is fit separately to each modeled variable.
- Hidden states are mapped to ordered regime labels using the training-state means.
- The resulting regime variables are used as nodes in a discrete Bayesian Network.

3.2 PCA-based pipeline

- A QuantileTransformer is fit on the training window and applied to validation data.
- PCA reduces grouped industrial, developing-country, and financial-market variables to two components per group.
- A three-state Gaussian HMM is fit to each processed variable and component.
- States are mapped to Low, Middle, and High using the training-state means, then passed to the Bayesian Network.

3.3 Bayesian Network learning and inference

Three structure-learning candidates are evaluated inside each rolling window: Hill Climbing with BIC, Hill Climbing with BDeu, and the PC algorithm. For each candidate, the network is fit on the training data and evaluated on the following validation period. The candidate with the highest validation Macro F1 is retained. Inference uses MAP prediction for the next-month regime.

Interpretation of edges

The learned directed edges are treated as conditional-dependency structure. They should not be interpreted as established causal effects without additional causal assumptions, identification strategy, or interventions.

3.4 Chronological validation design

The notebook uses non-anchored rolling windows with a 15-year training segment and a 2-year validation segment, while also allocating a forward 12-month segment in the window construction. The reported baseline/PCA comparison is based on training and validation metrics. A separate final unseen period from March 2023 through March 2024 is then used for the final regime prediction and trading experiment.

4. Rolling-Window Validation Results

Validation metric	Baseline	PCA pipeline	Difference
Accuracy	0.755	0.856	+0.101
Macro F1	0.710	0.724	+0.014
Weighted F1	0.753	0.844	+0.091
Micro F1	0.755	0.856	+0.101

The PCA-based pipeline has materially higher average validation accuracy, Micro F1, and Weighted F1. The improvement in Macro F1 is much smaller, indicating that the gain is concentrated in overall/majority-weighted classification rather than balanced performance across all three regimes. Training performance is also higher for the PCA pipeline, and the train-validation gap is larger, which is consistent with greater variance or overfitting risk.

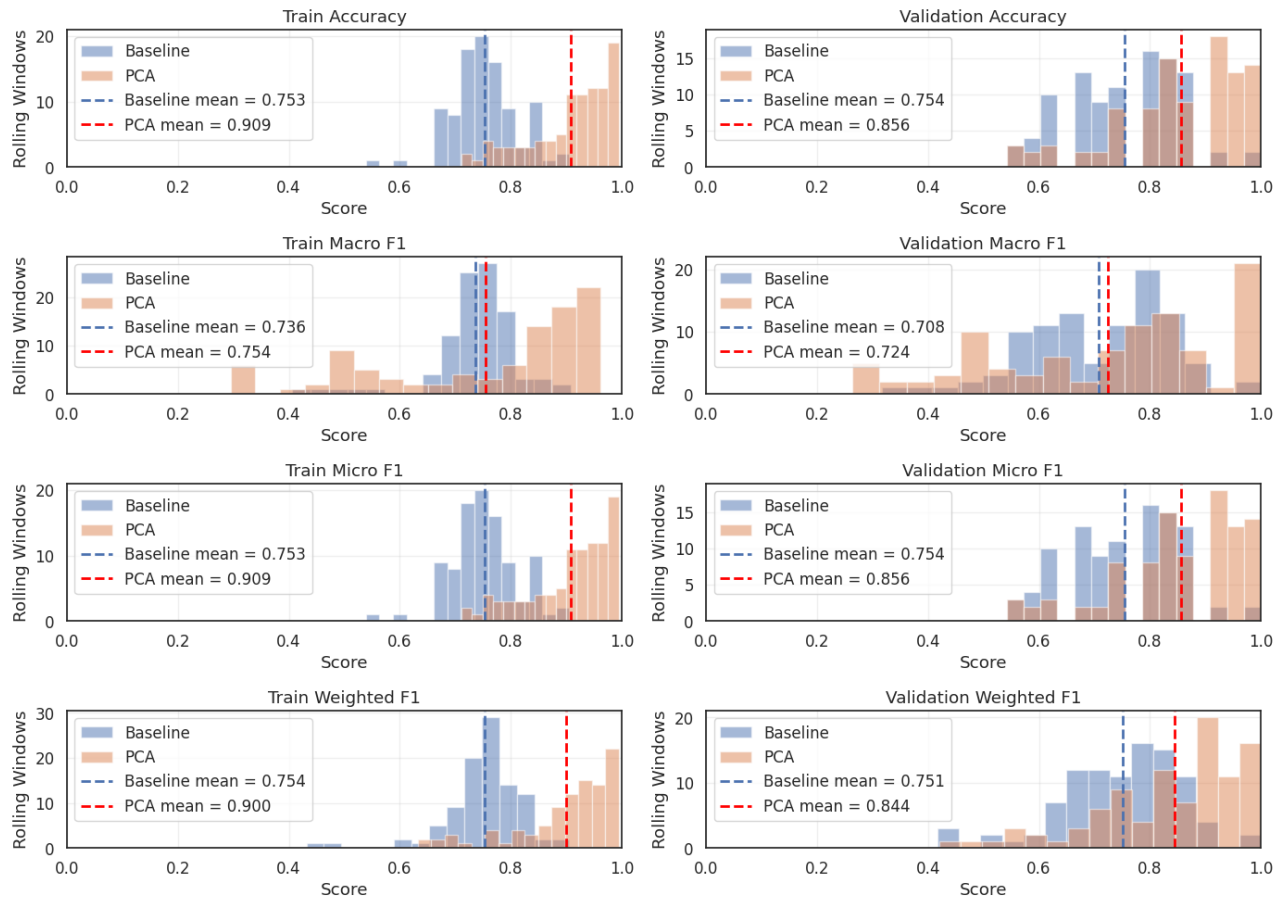


Figure 3. Distribution of training and validation metrics for the baseline and PCA-based pipelines.

Ablation caveat

The comparison should be described as baseline pipeline versus PCA-based pipeline—not as a clean causal test of PCA. The pipelines differ in more than dimensionality reduction: the baseline uses categorical HMMs after discretization, while the PCA pipeline uses Gaussian HMMs after quantile transformation and PCA. A controlled ablation using the same HMM/preprocessing with and without PCA would be required to isolate the incremental effect of PCA.

4.1 Model-selection frequency

Pipeline	Saved windows	HC-BIC	HC-BDeu	PC
Baseline	101	89 (88.1%)	11 (10.9%)	1 (1.0%)
PCA pipeline	97	86 (88.7%)	6 (6.2%)	5 (5.2%)

HC-BIC is the dominant selected structure-learning configuration in both pipelines. The saved notebook output contains 101 baseline windows and 97 PCA windows, so the aggregate summaries are not strictly paired over an identical set of windows. This should be standardized in a future refactor before making stronger head-to-head claims.

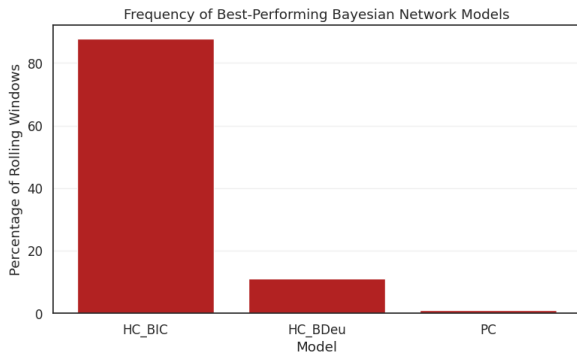


Figure 4a. Baseline BN selection frequency.

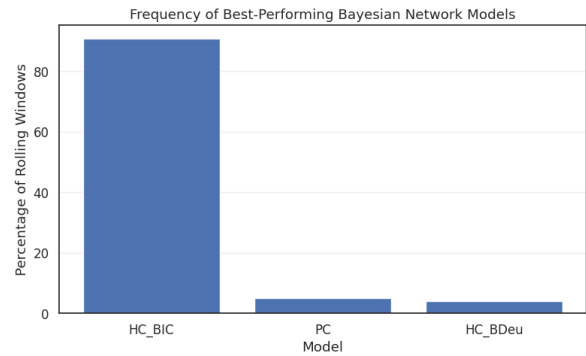


Figure 4b. PCA-pipeline BN selection frequency.

5. Network Structure and Stability

The network analysis asks two related questions: whether dimensionality reduction produces a more parsimonious graph, and whether the dependency structure remains stable through time.

Measure	Baseline	PCA pipeline
Mean number of edges	19.47	10.58
Edge-count standard deviation	5.35	3.76
Minimum edges	9	3
Maximum edges	37	25
Mean consecutive-network Jaccard similarity	0.198	0.155

The PCA-based pipeline produces substantially smaller Bayesian Networks, but lower mean Jaccard similarity indicates that exact edge configurations are not more stable from one rolling window to the next. Parsimony and structural stability are therefore separate properties.

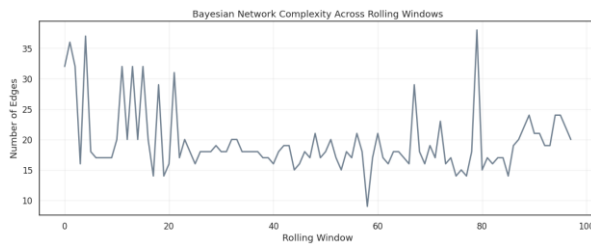


Figure 5a. Baseline network complexity.

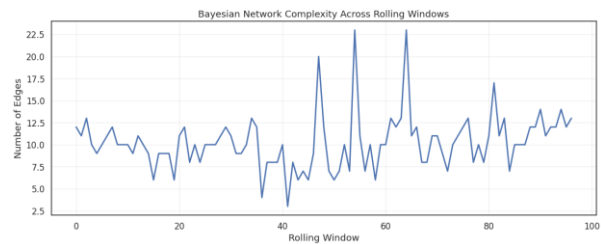


Figure 5b. PCA-pipeline network complexity.

5.1 What directly predicts the next-month oil regime?

The most persistent result is regime persistence. In the saved baseline output, the current crude-oil regime (cl_f) is a direct parent of the next-month target in all 101 rolling windows. In the PCA pipeline it appears in 92 of 97 windows (94.8%). Other industrial or market factors appear as direct target parents only occasionally.

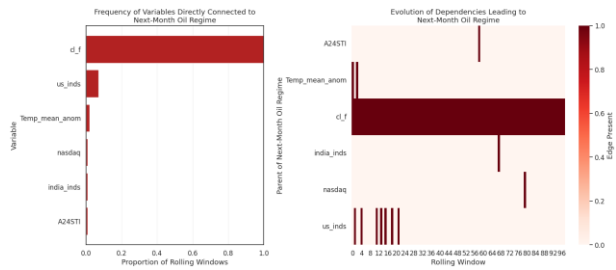


Figure 6a. Baseline target-parent frequency and timing.

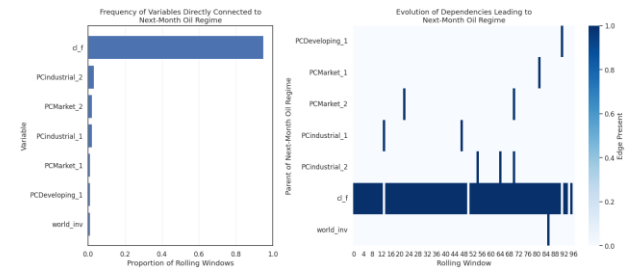


Figure 6b. PCA-pipeline target-parent frequency and timing.

Implication

The Bayesian Network is learning a strong first-order regime-persistence relationship. A simple persistence benchmark—predicting next month to remain in the current regime—should therefore be included in any future version of the project. Without that benchmark, the incremental predictive value of the full Bayesian Network cannot be isolated.

6. Final Unseen Period: March 2023–March 2024

After model selection, the PCA + HC-BIC framework is fit using the 15 years preceding March 2023 and evaluated on the subsequent unseen period. Twelve next-month predictions can be scored because the final March 2024 observation has no following-month target inside the dataset.

Final-test statistic	Result
Accuracy	1.00
Macro F1	1.00
Weighted F1	1.00
Micro F1	1.00
Predicted class in all 12 months	Middle
Actual class in all 12 months	Middle

Although the numerical scores are perfect, this final test does not contain Low or High target regimes. Consequently, it does not test whether the model can discriminate among all three classes under changing market conditions. The result is best interpreted as correct classification of a persistent Middle-regime episode, not as general evidence of 100% forecasting accuracy.

6.1 Trading experiment

The project maps predicted regimes to positions: High = +1 (long), Middle = 0 (flat), and Low = -1 (short).

Because every final-period prediction is Middle, the strategy remains flat in every month and generates a cumulative return of zero. Actual monthly crude-oil returns within the Middle regime still range roughly from -10.4% to +9.4%, showing that the three-state regime definition is too coarse to convert this particular test episode into differentiated trading signals.

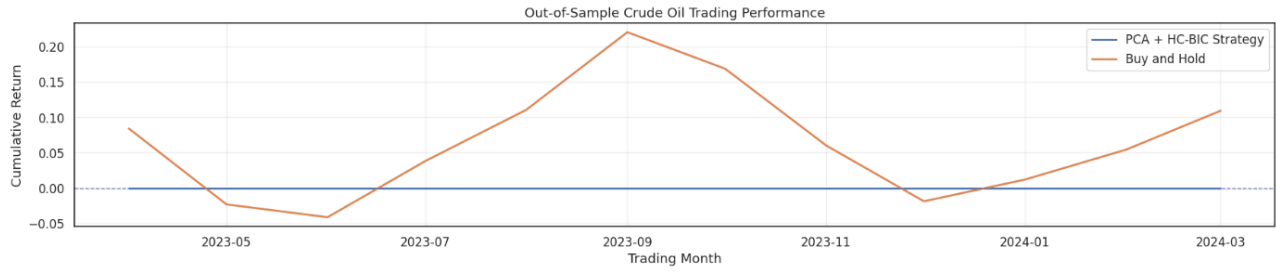


Figure 7. Out-of-sample trading performance: the strategy remains flat because all predicted regimes are Middle.

7. Temporal Evolution of the Bayesian Network

To examine structural change, the PCA + HC-BIC specification is estimated over three overlapping 15-year periods: pre-COVID (2004–2018), COVID-inclusive (2006–2020), and later period (2010–2024).

Period	Edges	Direct parent of current oil regime
Pre-COVID (2004–2018)	13	Temperature anomaly → current oil regime
COVID-inclusive (2006–2020)	13	Developing-country industrial factor → current oil regime
Later period (2010–2024)	9	Supply → current oil regime

The broader graph changes sharply across periods, while one forecasting relationship persists: the current crude-oil regime remains the only direct parent of the next-month regime in all three period-specific networks. Pairwise Jaccard similarities are very low (0.040, 0.100, and 0.048), indicating substantial reorganization of the surrounding dependency structure.

PCA + HC-BIC Bayesian Network
Pre-COVID (2004-2018)

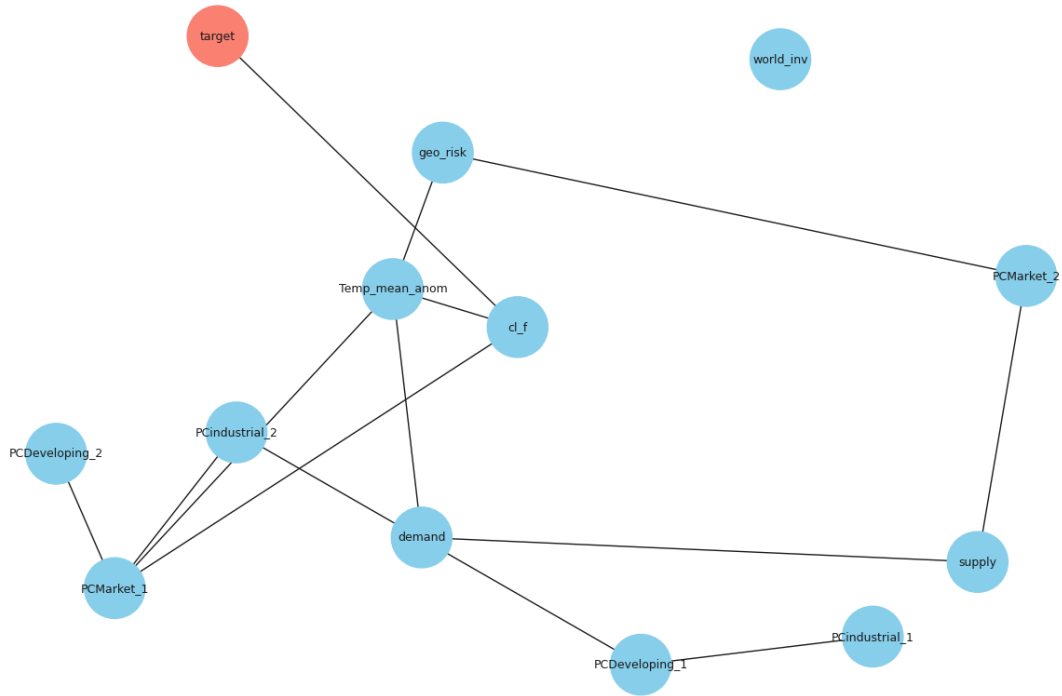


Figure 8. PCA + HC-BIC Bayesian Network, pre-COVID period (2004–2018).

PCA + HC-BIC Bayesian Network
COVID-inclusive (2006-2020)

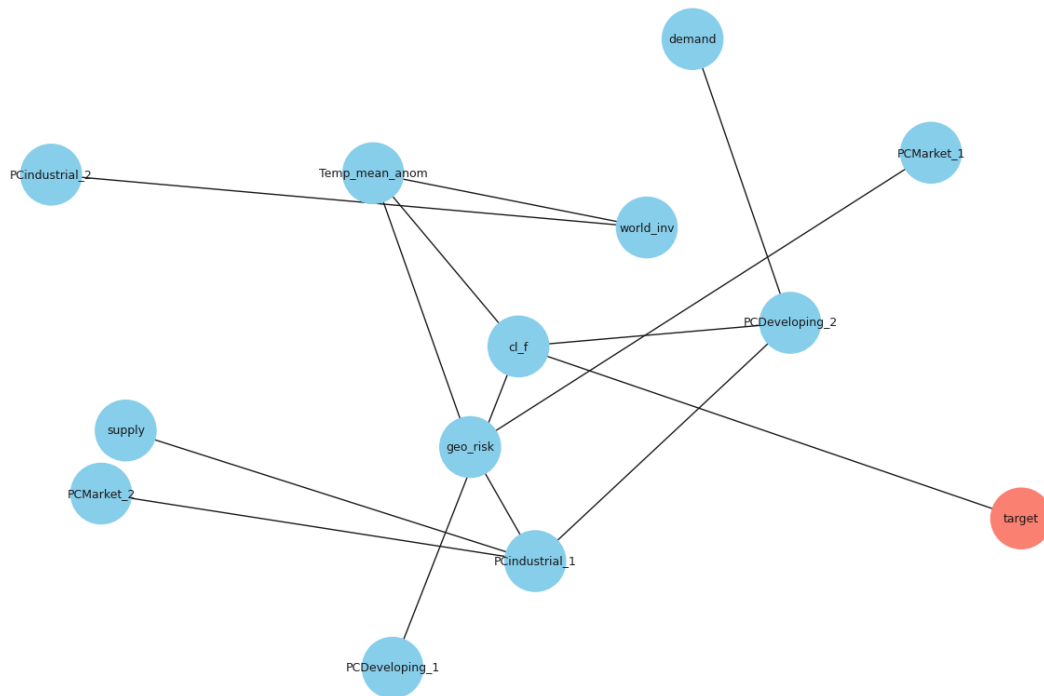


Figure 9. PCA + HC-BIC Bayesian Network, COVID-inclusive period (2006–2020).

PCA + HC-BIC Bayesian Network
Later period (2010-2024)

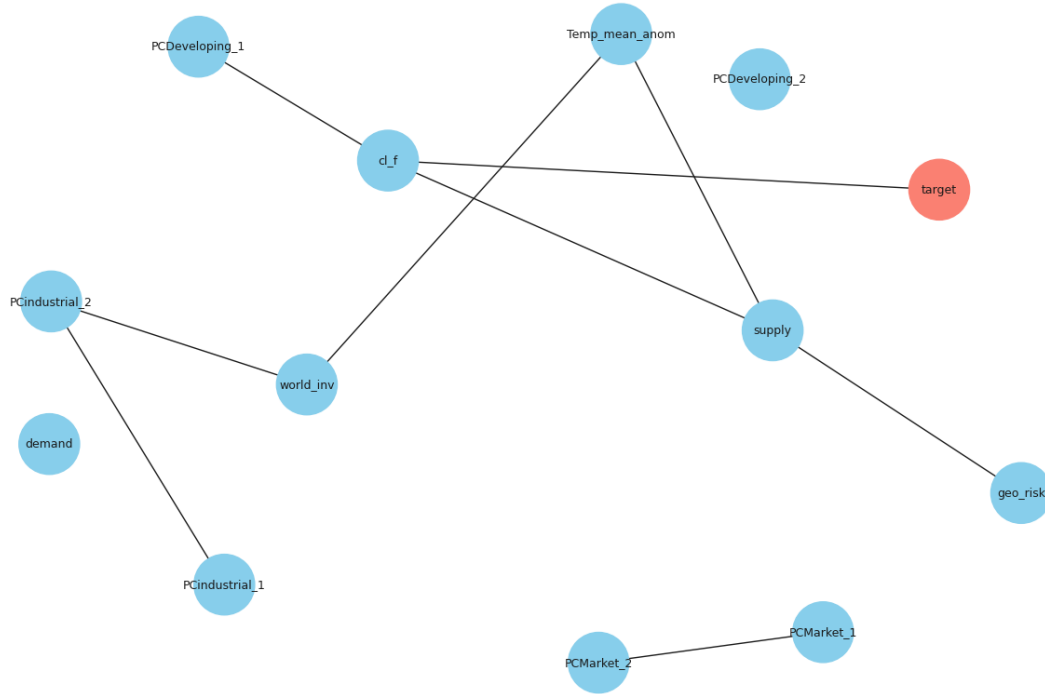


Figure 10. PCA + HC-BIC Bayesian Network, later period (2010–2024).

8. Interpretation, Limitations, and Next Steps

The project demonstrates a technically interesting combination of regime detection, probabilistic graphical modeling, dimensionality reduction, and chronological validation. Its strongest result is not that Bayesian Networks “solve” crude-oil forecasting, but that the market can be represented as a changing dependency system in which regime persistence is highly stable while the surrounding macro-financial structure is time-varying.

Recommended next iteration

- Add a persistence/Markov-transition benchmark and conventional classifiers such as multinomial logistic regression or gradient boosting.
- Use identical rolling dates for all candidate pipelines and report paired differences with confidence intervals or bootstrap uncertainty.
- Run a clean PCA ablation: same preprocessing, same Gaussian HMM, same BN procedure, with and without PCA.
- Report per-class precision/recall/F1 and confusion matrices in addition to aggregate metrics.
- Use posterior regime probabilities, not only MAP classes, and evaluate probabilistic scores such as log loss or Brier score.
- Redesign the trading layer using regime probabilities, return forecasts, or more granular states, then include transaction costs and risk-adjusted benchmarks.
- Reserve a longer final test period containing multiple regime transitions.

9. Conclusion

The PCA-based HMM–Bayesian Network pipeline achieves stronger average validation accuracy and weighted F1 than the categorical baseline while producing a substantially more compact network. However, Macro F1 improves only modestly and the PCA pipeline is more variable. Across rolling windows and period-

specific networks, the most stable relationship is the transition from the current crude-oil regime to the next-month regime.

The final unseen test correctly classifies all 12 observations, but every realized target is Middle. The corresponding trading rule therefore remains flat and earns zero cumulative return. This makes the project's main contribution methodological and interpretive rather than a demonstrated profitable trading strategy: it provides a structured framework for combining market regimes, macro-financial information, and probabilistic dependencies while making the limitations of the current predictive evidence explicit.

10. Reproducibility and Technology Stack

Component	Tools
Language	Python
Core data tools	pandas, NumPy
Time series / diagnostics	statsmodels, arch
Regime models	hmmlearn
Bayesian Networks	pgmpy
Machine learning / PCA	scikit-learn
Graph analysis	NetworkX
Visualization	Matplotlib
Market data	Yahoo Finance / yfinance

Selected References and Data Sources

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- U.S. Energy Information Administration. Short-Term Energy Outlook / Open Data.
- Yahoo Finance. Historical Market Data.
- pgmpy documentation and source repository.
- hmmlearn documentation.

This document is a portfolio summary of the submitted project materials and saved notebook outputs. It is not investment advice.